Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Forebay Farms

Physical address of dairy:

13699 Cherokee RD	Gustine	Merced	95322
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1910

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X078-X130-X053-XXXX X078-X140-X074-XXXX X078-X140-X075-XXXX X078-X140-X076-XXXX X078-X140-X077-XXXX X078-X140-X078-XXXX

B. OPERATORS

C/O Ralph, Lloyd, &/or Fred Fagundes, Forebay Farms

Operator name: C/O Ralph, Lloyd, &/or Fred Fagundes, Forebay Farms Telephone no.: (209) 383-6046 Landline Cellular

P.O. Box 2717	Merced	CA	95344
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

C/O Ralph, Lloyd, &/or Fred Fagundes, Forebay Farms

Legal owner name: C/O Ralph, Lloyd, &/or Fred Fagundes, Forebay Farms Telephone no.: (209) 383-6046 Landline Cellular

P.O. Box 2717	Merced	CA	95344
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	1,223	102	0	0	0	0
Maximum number	1,223	102	0	0	0	0
Average number	1,223	102	0	0	0	0
Avg live weight (lbs)	1,223	102	0	0		

Predominant milk cow breed: Holstein

Average milk production: 60 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 30,865.92 *tons per reporting period*

Total nitrogen from manure: 410,018.42 *lbs per reporting period*

After ammonia losses (30% loss applied): 287,012.89 *lbs per reporting period*

Total phosphorus from manure: 68,445.84 *lbs per reporting period*

Total potassium from manure: 226,180.67 *lbs per reporting period*

Total salt from manure: 599,304.45 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 30,748,125 *gallons*

Total nitrogen generated: 29,906.14 *lbs*

Total phosphorus generated: 5,771.95 *lbs*

Total potassium generated: 38,451.56 *lbs*

Total salt generated: 524,386.03 *lbs*

	30,748,125	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	30,748,125	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - DW 1	Ground water
Domestic Well - DW 2	Ground water
Domestic Well - DW 3	Ground water
Irrigation Water - Surface Water	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	27	27	0	none	X078-X140-X074-XXXX
10	31	31	2	both	X078-X140-X005-XXXX
11A	51	51	2	both	X078-X140-X005-XXXX
11B	52	52	2	both	X078-X140-X005-XXXX
12	13	13	0	none	X078-X140-X017-XXXX
2	25	25	0	none	X078-X130-X053-XXXX
3	52	52	0	none	X078-X130-X053-XXXX
4	20	20	0	none	X078-X130-X053-XXXX
5	42	42	0	none	X078-X140-X011-XXXX
6	40	40	0	none	X078-X140-X011-XXXX
7	30	30	2	both	X078-X140-X078-XXXX
8	70	70	2	both	X078-X140-X078-XXXX
9	14	14	0	none	X078-X140-X078-XXXX
Totals for areas that were used for application	234	234	10		
Totals for areas that were not used for application	233	233	0		
Land application area totals	467	467	10		

B. CROPS AND HARVESTS

10

Field name: 10

12/03/2022: Ryegrass, hay

Crop: Ryegrass, hay Acres planted: 31 Plant date: 12/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	278.00 *ton*	Dry-weight		78.4	24,700.00	4,500.00	43,400.00		13.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	192.00	27.60	252.00	0.00
Total actual harvest content	8.97	95.69	17.43	168.13	534.62

10

05/10/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 31 Plant date: 05/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
07/20/2023	362.00 *ton*	Dry-weight		74.9	14,400.00	2,500.00	20,800.00		7.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	11.68	84.41	14.66	121.93	433.79

11A

Field name: 11A

12/03/2022: Ryegrass, hay

Crop: Ryegrass, hay Acres planted: 51 Plant date: 12/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	453.00 *ton*	Dry-weight		76.3	19,600.00	4,400.00	39,000.00		13.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	192.00	27.60	252.00	0.00
Total actual harvest content	8.88	82.52	18.53	164.20	576.80

05/18/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 51 Plant date: 05/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	581.00 *ton*	Dry-weight		71.7	13,900.00	2,500.00	26,700.00		11.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	11.39	89.63	16.12	172.16	715.72

11B

Field name: 11B

12/03/2022: Ryegrass, hay

Crop: Ryegrass, hay Acres planted: 52 Plant date: 12/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	467.00 *ton*	Dry-weight		77.2	25,300.00	4,600.00	43,200.00		13.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	192.00	27.60	252.00	0.00
Total actual harvest content	8.98	103.61	18.84	176.91	556.95

05/10/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 52 Plant date: 05/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
07/20/2023	608.00 *ton*	Dry-weight		77.4	15,500.00	2,800.00	28,100.00		12.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	11.69	81.92	14.80	148.51	660.62

7

Field name: 7

12/03/2022: Ryegrass, hay

Crop: Ryegrass, hay Acres planted: 30 Plant date: 12/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	268.00 *ton*	Dry-weight		75.5	24,100.00	4,500.00	40,400.00		14.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	192.00	27.60	252.00	0.00
Total actual harvest content	8.93	105.49	19.70	176.84	625.96

7

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	898.00 *ton*	Dry-weight		71.6	13,600.00	2,500.00	17,100.00		7.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.93	231.23	42.51	290.74	1,190.15

8

Field name: 8

12/03/2022: Ryegrass, hay

Crop: Ryegrass, hay Acres planted: 70 Plant date: 12/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	587.00 *ton*	Dry-weight		76.7	27,000.00	4,600.00	42,500.00		13.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	192.00	27.60	252.00	0.00
Total actual harvest content	8.39	105.51	17.98	166.08	515.82

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 70 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	2,250.00 *ton*	Dry-weight		71.7	11,800.00	2,500.00	17,400.00		7.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.14	214.68	45.48	316.56	1,273.50

NUTRIENT BUDGET

A. LAND APPLICATIONS

10 - 12/03/2022: Ryegrass, hay

Field name: 10

Crop: Ryegrass, hay

Plant date: 12/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	24.50	4.54	30.08	390.48	653,400.00 *gal*
Irrigation Water - Surface Water	Surface water	0.29	0.00	0.00	163.78	3,537,222.00 *gal*
Application event totals		24.79	4.54	30.08	554.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	23.94	4.43	29.39	381.60	638,550.00 *gal*
Irrigation Water - Surface Water	Surface water	0.28	0.00	0.00	159.05	3,435,187.00 *gal*
Application event totals		24.22	4.43	29.39	540.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	10.22	2.58	17.71	315.66	651,450.00 *gal*
Irrigation Water - Surface Water	Surface water	0.26	0.00	0.00	149.73	3,233,810.00 *gal*
Application event totals		10.49	2.58	17.71	465.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/04/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	66.80	19.28	57.16	0.00	250.00 *ton*
Application event totals		66.80	19.28	57.16	0.00	

10 - 05/10/2023: Sudangrass, silage

10 - 05/10/2023: Sudangrass, silage

Field name: 10

Crop: Sudangrass, silage

Plant date: 05/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	99.77	20.75	214.37	0.00	240.00 *ton*
Application event totals		99.77	20.75	214.37	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	2.75	0.93	6.43	107.67	499,950.00 *gal*
Irrigation Water - Surface Water	Surface water	0.27	0.00	0.00	154.36	3,333,825.00 *gal*
Application event totals		3.01	0.93	6.43	262.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.25	0.00	0.00	145.10	3,133,796.00 *gal*
Application event totals		0.25	0.00	0.00	145.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	2.66	0.90	6.24	104.40	484,800.00 *gal*
Irrigation Water - Surface Water	Surface water	0.22	0.00	0.00	125.03	2,700,398.00 *gal*
Application event totals		2.88	0.90	6.24	229.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	2.79	0.95	6.53	109.30	507,525.00 *gal*
Irrigation Water - Surface Water	Surface water	0.24	0.00	0.00	135.84	2,933,766.00 *gal*
Application event totals		3.02	0.95	6.53	245.14	

10 - 05/10/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.22	0.00	0.00	126.58	2,733,737.00 *gal*
Application event totals		0.22	0.00	0.00	126.58	

11A - 12/03/2022: Ryegrass, hay

Field name: 11A

Crop: Ryegrass, hay

Plant date: 12/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	10.62	2.68	18.40	327.97	1,113,525.00 *gal*
Irrigation Water - Surface Water	Surface water	0.29	0.00	0.00	167.51	5,952,056.00 *gal*
Application event totals		10.91	2.68	18.40	495.48	

02/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	10.48	2.64	18.15	323.50	1,098,375.00 *gal*
Irrigation Water - Surface Water	Surface water	0.29	0.00	0.00	164.64	5,850,021.00 *gal*
Application event totals		10.77	2.64	18.15	488.15	

03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	24.71	4.58	30.33	393.78	1,084,050.00 *gal*
Irrigation Water - Surface Water	Surface water	0.29	0.00	0.00	168.89	6,000,885.00 *gal*
Application event totals		25.00	4.58	30.33	562.67	

11A - 12/03/2022: Ryegrass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/04/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	60.65	12.61	130.30	0.00	240.00 *ton*
Application event totals		60.65	12.61	130.30	0.00	

11A - 05/18/2023: Sudangrass, silage

Field name: 11A

Crop: Sudangrass, silage

Plant date: 05/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	53.07	11.03	114.01	0.00	210.00 *ton*
Application event totals		53.07	11.03	114.01	0.00	

05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.21	0.00	0.00	118.67	4,216,406.00 *gal*
Application event totals		0.21	0.00	0.00	118.67	

05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.65	3.82	25.35	329.05	905,850.00 *gal*
Irrigation Water - Surface Water	Surface water	0.20	0.00	0.00	116.90	4,153,676.00 *gal*
Application event totals		20.85	3.82	25.35	445.95	

06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.20	0.00	0.00	113.12	4,019,252.00 *gal*
Application event totals		0.20	0.00	0.00	113.12	

11A - 05/18/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.82	3.86	25.55	331.75	913,275.00 *gal*
Irrigation Water - Surface Water	Surface water	0.19	0.00	0.00	108.57	3,857,559.00 *gal*
Application event totals		21.01	3.86	25.55	440.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.19	0.00	0.00	110.76	3,935,512.00 *gal*
Application event totals		0.19	0.00	0.00	110.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.65	3.82	25.35	329.05	905,850.00 *gal*
Irrigation Water - Surface Water	Surface water	0.20	0.00	0.00	112.53	3,998,210.00 *gal*
Application event totals		20.84	3.82	25.35	441.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.19	0.00	0.00	110.76	3,935,507.00 *gal*
Application event totals		0.19	0.00	0.00	110.76	

11B - 12/03/2022: Ryegrass, hay

Field name: 11B

Crop: Ryegrass, hay

Plant date: 12/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

11B - 12/03/2022: Ryegrass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.57	4.18	27.71	359.76	1,009,800.00 *gal*
Irrigation Water - Surface Water	Surface water	0.26	0.00	0.00	150.92	5,467,473.00 *gal*
Application event totals		22.84	4.18	27.71	510.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.24	4.12	27.30	354.47	994,950.00 *gal*
Irrigation Water - Surface Water	Surface water	0.26	0.00	0.00	148.16	5,367,458.00 *gal*
Application event totals		22.50	4.12	27.30	502.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	9.57	2.41	16.58	295.40	1,022,625.00 *gal*
Irrigation Water - Surface Water	Surface water	0.28	0.00	0.00	158.66	5,747,986.00 *gal*
Application event totals		9.84	2.41	16.58	454.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/04/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	69.39	14.43	149.10	0.00	280.00 *ton*
Application event totals		69.39	14.43	149.10	0.00	

11B - 05/10/2023: Sudangrass, silage

Field name: 11B

Crop: Sudangrass, silage

Plant date: 05/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

11B - 05/10/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	99.14	20.61	212.99	0.00	400.00 *ton*
Application event totals		99.14	20.61	212.99	0.00	

05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.24	0.00	0.00	138.94	5,033,739.00 *gal*
Application event totals		0.24	0.00	0.00	138.94	

05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.25	0.00	0.00	140.82	5,101,763.00 *gal*
Application event totals		0.25	0.00	0.00	140.82	

06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.24	0.00	0.00	138.95	5,034,076.00 *gal*
Application event totals		0.24	0.00	0.00	138.95	

06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.24	0.00	0.00	138.95	5,034,076.00 *gal*
Application event totals		0.24	0.00	0.00	138.95	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.24	0.00	0.00	135.27	4,900,723.00 *gal*
Application event totals		0.24	0.00	0.00	135.27	

7 - 12/03/2022: Ryegrass, hay

7 - 12/03/2022: Ryegrass, hay

Field name: 7

Crop: Ryegrass, hay

Plant date: 12/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	4.21	1.43	9.87	165.20	742,350.00 *gal*
Irrigation Water - Surface Water	Surface water	0.31	0.00	0.00	177.05	3,700,546.00 *gal*
Application event totals		4.52	1.43	9.87	342.25	

02/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	12.16	3.07	21.07	375.49	749,925.00 *gal*
Irrigation Water - Surface Water	Surface water	0.32	0.00	0.00	185.03	3,867,237.00 *gal*
Application event totals		12.48	3.07	21.07	560.51	

03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	11.67	2.94	20.22	360.32	719,625.00 *gal*
Irrigation Water - Surface Water	Surface water	0.38	0.00	0.00	216.93	4,534,002.00 *gal*
Application event totals		12.05	2.94	20.22	577.24	

11/04/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	110.45	31.88	94.51	0.00	400.00 *ton*
Application event totals		110.45	31.88	94.51	0.00	

7 - 05/31/2023: Corn, silage

Field name: 7

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

7 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	179.48	51.81	153.58	0.00	650.00 *ton*
Application event totals		179.48	51.81	153.58	0.00	

06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	24.30	4.50	29.82	387.19	627,000.00 *gal*
Irrigation Water - Surface Water	Surface water	0.40	0.00	0.00	228.09	4,767,370.00 *gal*
Application event totals		24.69	4.50	29.82	615.28	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.39	0.00	0.00	223.31	4,667,355.00 *gal*
Application event totals		0.39	0.00	0.00	223.31	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	24.57	4.55	30.16	391.59	634,125.00 *gal*
Irrigation Water - Surface Water	Surface water	0.35	0.00	0.00	202.57	4,233,958.00 *gal*
Application event totals		24.92	4.55	30.16	594.16	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.42	4.89	32.43	421.00	681,750.00 *gal*
Irrigation Water - Surface Water	Surface water	0.35	0.00	0.00	201.78	4,217,457.00 *gal*
Application event totals		26.77	4.89	32.43	622.78	

7 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	25.54	4.73	31.35	406.97	659,025.00 *gal*
Irrigation Water - Surface Water	Surface water	0.33	0.00	0.00	190.39	3,979,375.00 *gal*
Application event totals		25.87	4.73	31.35	597.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	25.83	4.78	31.71	411.65	666,600.00 *gal*
Irrigation Water - Surface Water	Surface water	0.32	0.00	0.00	182.26	3,809,316.00 *gal*
Application event totals		26.15	4.78	31.71	593.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.33	0.00	0.00	188.76	3,945,363.00 *gal*
Application event totals		0.33	0.00	0.00	188.76	

8 - 12/03/2022: Ryegrass, hay

Field name: 8

Crop: Ryegrass, hay

Plant date: 12/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	25.40	4.70	31.18	404.80	1,529,550.00 *gal*
Irrigation Water - Surface Water	Surface water	0.28	0.00	0.00	158.59	7,734,474.00 *gal*
Application event totals		25.68	4.70	31.18	563.40	

8 - 12/03/2022: Ryegrass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.76	4.96	32.85	426.42	1,611,225.00 *gal*
Irrigation Water - Surface Water	Surface water	0.28	0.00	0.00	157.91	7,701,136.00 *gal*
Application event totals		27.03	4.96	32.85	584.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	27.67	5.13	33.97	441.05	1,666,500.00 *gal*
Irrigation Water - Surface Water	Surface water	0.38	0.00	0.00	216.02	10,534,887.00 *gal*
Application event totals		28.05	5.13	33.97	657.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/04/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	58.91	12.25	126.58	0.00	320.00 *ton*
Application event totals		58.91	12.25	126.58	0.00	

8 - 05/31/2023: Corn, silage

Field name: 8

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	156.49	32.54	336.23	0.00	850.00 *ton*
Application event totals		156.49	32.54	336.23	0.00	

8 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.17	4.85	32.12	416.99	1,575,600.00 *gal*
Irrigation Water - Surface Water	Surface water	0.37	0.00	0.00	209.92	10,237,537.00 *gal*
Application event totals		26.53	4.85	32.12	626.91	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.37	0.00	0.00	212.01	10,339,572.00 *gal*
Application event totals		0.37	0.00	0.00	212.01	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.92	4.99	33.05	429.02	1,621,050.00 *gal*
Irrigation Water - Surface Water	Surface water	0.34	0.00	0.00	196.67	9,591,314.00 *gal*
Application event totals		27.26	4.99	33.05	625.69	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.54	4.92	32.58	423.01	1,598,325.00 *gal*
Irrigation Water - Surface Water	Surface water	0.34	0.00	0.00	194.58	9,489,278.00 *gal*
Application event totals		26.88	4.92	32.58	617.58	

08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.42	4.89	32.43	421.00	1,590,750.00 *gal*
Irrigation Water - Surface Water	Surface water	0.34	0.00	0.00	195.51	9,534,740.00 *gal*
Application event totals		26.76	4.89	32.43	616.51	

8 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.42	4.89	32.43	421.00	1,590,750.00 *gal*
Irrigation Water - Surface Water	Surface water	0.34	0.00	0.00	193.46	9,434,725.00 *gal*
Application event totals		26.75	4.89	32.43	614.46	

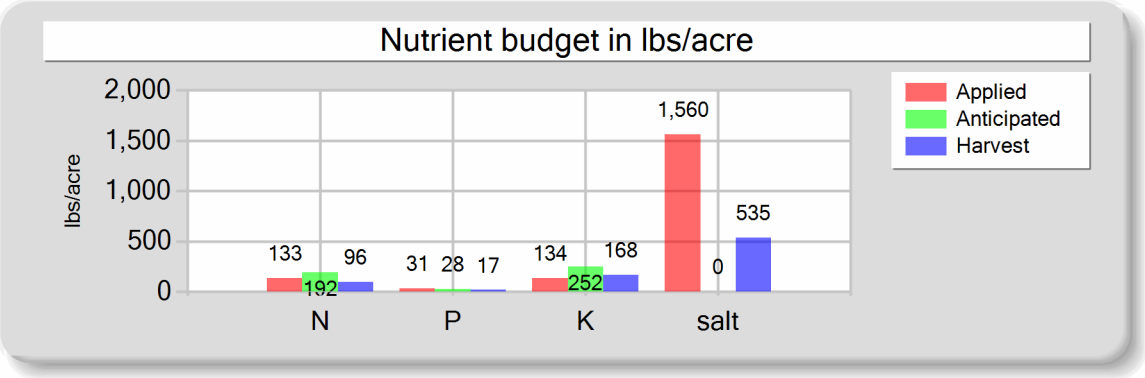
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.34	0.00	0.00	194.14	9,468,063.00 *gal*
Application event totals		0.34	0.00	0.00	194.14	

B. NUTRIENT BUDGET

10 - 12/03/2022: Ryegrass, hay

Field name: 10 Crop: Ryegrass, hay Plant date: 12/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	66.80	19.28	57.16	0.00
Process wastewater	58.67	11.55	77.18	1,087.74
Fresh water	0.82	0.00	0.00	472.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	133.30	30.83	134.35	1,560.30
Anticipated crop nutrient removal	192.00	27.60	252.00	0.00
Actual crop nutrient removal	95.69	17.43	168.13	534.62
Nutrient balance	37.61	13.40	-33.79	1,025.68
Applied to removed ratio	1.39	1.77	0.80	2.92

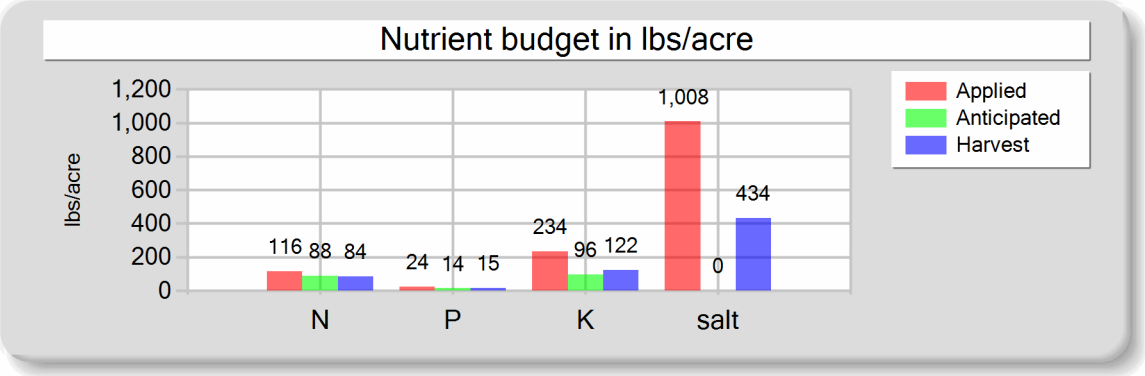
Fresh water applied	
10,206,219.00	*gallons*
375.86	*acre-inches*
12.12	*inches/acre*

Process wastewater applied	
1,943,400.00	*gallons*
71.57	*acre-inches*
2.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 - 05/10/2023: Sudangrass, silage

Field name: 10 Crop: Sudangrass, silage Plant date: 05/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	99.77	20.75	214.37	0.00
Process wastewater	8.19	2.78	19.20	321.37
Fresh water	1.20	0.00	0.00	686.90
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	116.17	23.53	233.57	1,008.27
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	84.41	14.66	121.93	433.79
Nutrient balance	31.75	8.87	111.64	574.48
Applied to removed ratio	1.38	1.61	1.92	2.32

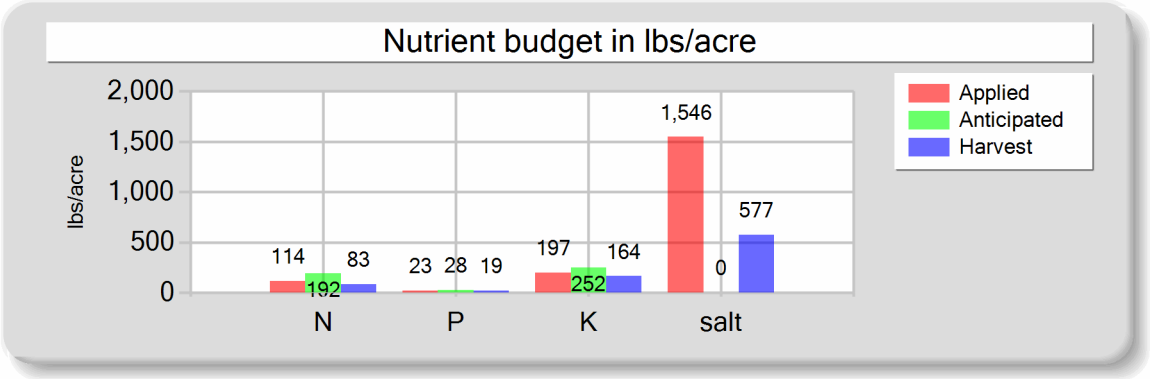
Fresh water applied	
14,835,522.00	*gallons*
546.34	*acre-inches*
17.62	*inches/acre*

Process wastewater applied	
1,492,275.00	*gallons*
54.96	*acre-inches*
1.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

11A - 12/03/2022: Ryegrass, hay

Field name: 11A Crop: Ryegrass, hay Plant date: 12/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	60.65	12.61	130.30	0.00
Process wastewater	45.81	9.90	66.89	1,045.25
Fresh water	0.87	0.00	0.00	501.05
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	114.33	22.51	197.19	1,546.30
Anticipated crop nutrient removal	192.00	27.60	252.00	0.00
Actual crop nutrient removal	82.52	18.53	164.20	576.80
Nutrient balance	31.81	3.98	32.99	969.50
Applied to removed ratio	1.39	1.21	1.20	2.68

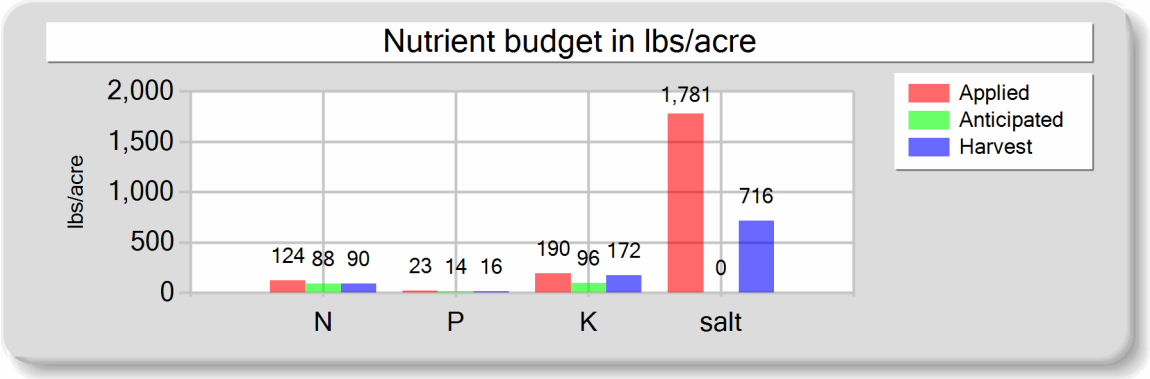
Fresh water applied	
17,802,962.00	*gallons*
655.62	*acre-inches*
12.86	*inches/acre*

Process wastewater applied	
3,295,950.00	*gallons*
121.38	*acre-inches*
2.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

11A - 05/18/2023: Sudangrass, silage

Field name: 11A | Crop: Sudangrass, silage | Plant date: 05/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	53.07	11.03	114.01	0.00
Process wastewater	62.11	11.50	76.25	989.86
Fresh water	1.38	0.00	0.00	791.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	123.56	22.54	190.26	1,781.15
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	89.63	16.12	172.16	715.72
Nutrient balance	33.93	6.42	18.10	1,065.43
Applied to removed ratio	1.38	1.40	1.11	2.49

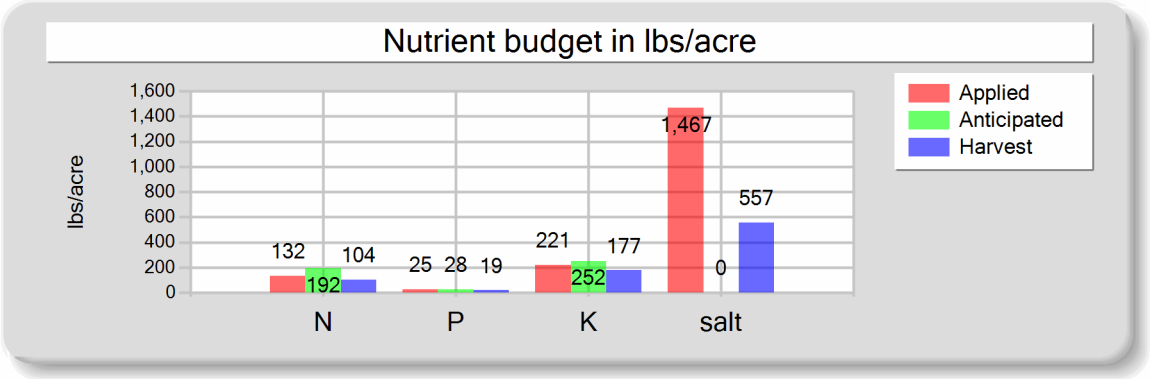
Fresh water applied	
28,116,122.00	*gallons*
1,035.42	*acre-inches*
20.30	*inches/acre*

Process wastewater applied	
2,724,975.00	*gallons*
100.35	*acre-inches*
1.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

11B - 12/03/2022: Ryegrass, hay

Field name: 11B Crop: Ryegrass, hay Plant date: 12/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	69.39	14.43	149.10	0.00
Process wastewater	54.38	10.71	71.59	1,009.63
Fresh water	0.80	0.00	0.00	457.73
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	131.58	25.14	220.69	1,467.36
Anticipated crop nutrient removal	192.00	27.60	252.00	0.00
Actual crop nutrient removal	103.61	18.84	176.91	556.95
Nutrient balance	27.97	6.30	43.77	910.41
Applied to removed ratio	1.27	1.33	1.25	2.63

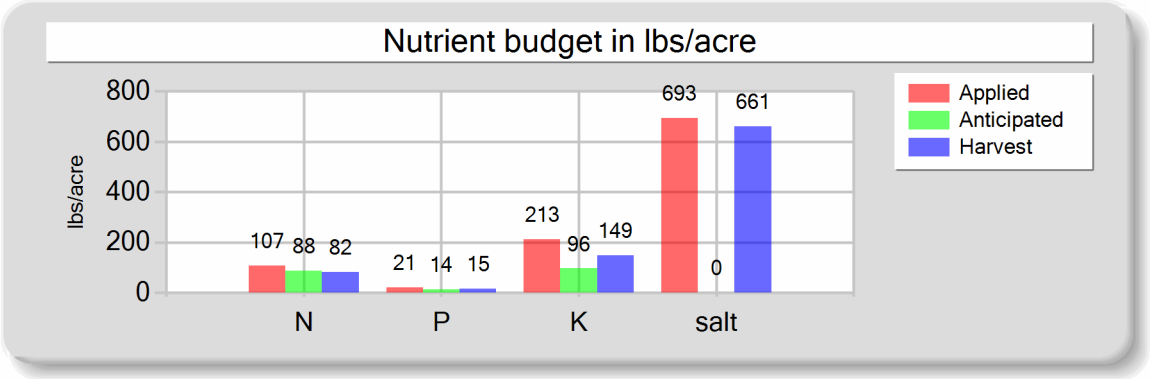
Fresh water applied	
16,582,917.00	*gallons*
610.69	*acre-inches*
11.74	*inches/acre*

Process wastewater applied	
3,027,375.00	*gallons*
111.49	*acre-inches*
2.14	*inches/acre*

Total harvests for the crop	
1	*harvests*

11B - 05/10/2023: Sudangrass, silage

Field name: 11B Crop: Sudangrass, silage Plant date: 05/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	99.14	20.61	212.99	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.21	0.00	0.00	692.95
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	107.34	20.61	212.99	692.95
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	81.92	14.80	148.51	660.62
Nutrient balance	25.43	5.81	64.49	32.33
Applied to removed ratio	1.31	1.39	1.43	1.05

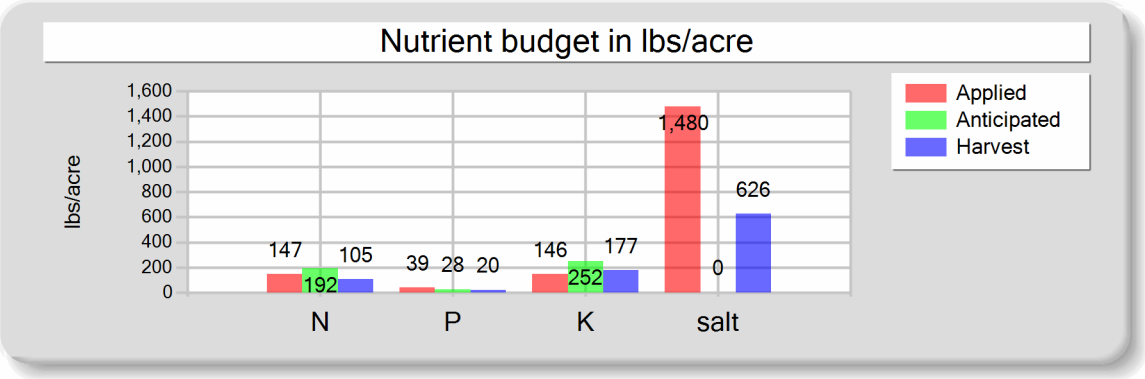
Fresh water applied	
25,104,377.00	*gallons*
924.51	*acre-inches*
17.78	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 12/03/2022: Ryegrass, hay

Field name: 7 Crop: Ryegrass, hay Plant date: 12/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	110.45	31.88	94.51	0.00
Process wastewater	28.04	7.44	51.16	901.00
Fresh water	1.01	0.00	0.00	579.01
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	146.50	39.32	145.67	1,480.01
Anticipated crop nutrient removal	192.00	27.60	252.00	0.00
Actual crop nutrient removal	105.49	19.70	176.84	625.96
Nutrient balance	41.01	19.62	-31.18	854.05
Applied to removed ratio	1.39	2.00	0.82	2.36

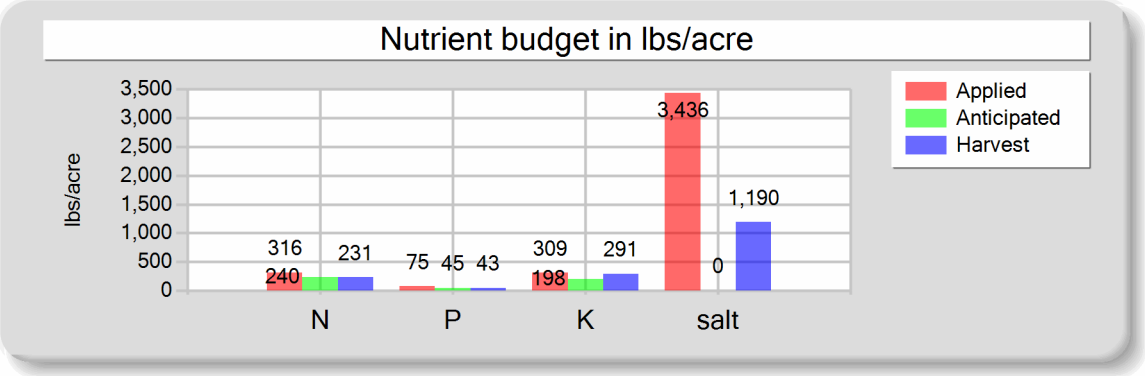
Fresh water applied	
12,101,785.00	*gallons*
445.67	*acre-inches*
14.86	*inches/acre*

Process wastewater applied	
2,211,900.00	*gallons*
81.46	*acre-inches*
2.72	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 05/31/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	179.48	51.81	153.58	0.00
Process wastewater	126.65	23.46	155.47	2,018.40
Fresh water	2.47	0.00	0.00	1,417.17
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	315.60	75.27	309.05	3,435.57
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	231.23	42.51	290.74	1,190.15
Nutrient balance	84.37	32.76	18.31	2,245.42
Applied to removed ratio	1.36	1.77	1.06	2.89

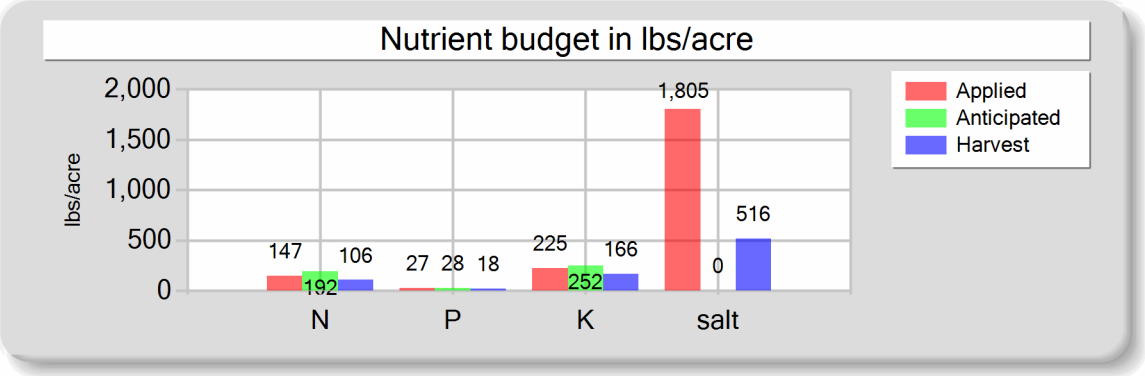
Fresh water applied	
29,620,194.00	*gallons*
1,090.81	*acre-inches*
36.36	*inches/acre*

Process wastewater applied	
3,268,500.00	*gallons*
120.37	*acre-inches*
4.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 12/03/2022: Ryegrass, hay

Field name: 8 Crop: Ryegrass, hay Plant date: 12/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	58.91	12.25	126.58	0.00
Process wastewater	79.83	14.79	98.00	1,272.27
Fresh water	0.93	0.00	0.00	532.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	146.68	27.04	224.58	1,804.79
Anticipated crop nutrient removal	192.00	27.60	252.00	0.00
Actual crop nutrient removal	105.51	17.98	166.08	515.82
Nutrient balance	41.17	9.06	58.50	1,288.97
Applied to removed ratio	1.39	1.50	1.35	3.50

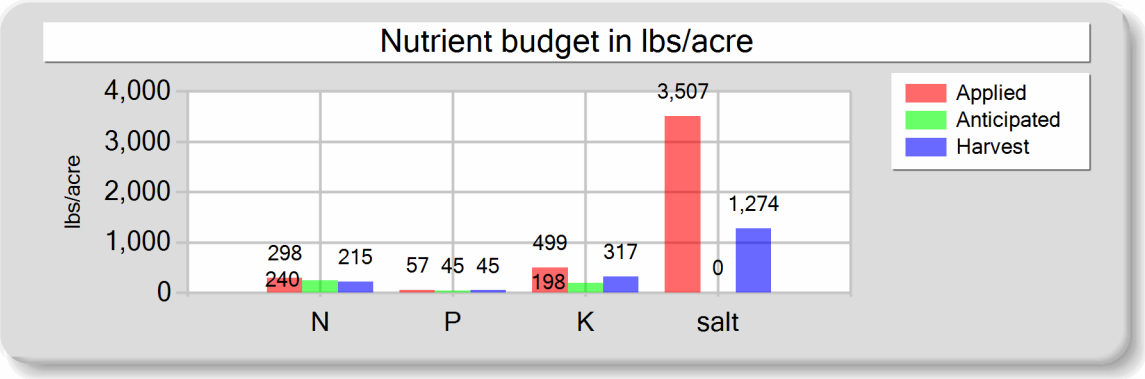
Fresh water applied	
25,970,497.00	*gallons*
956.41	*acre-inches*
13.66	*inches/acre*

Process wastewater applied	
4,807,275.00	*gallons*
177.04	*acre-inches*
2.53	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 05/31/2023: Corn, silage

Field name: 8 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	156.49	32.54	336.23	0.00
Process wastewater	132.46	24.53	162.61	2,111.02
Fresh water	2.44	0.00	0.00	1,396.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	298.39	57.07	498.83	3,507.30
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	214.68	45.48	316.56	1,273.50
Nutrient balance	83.71	11.59	182.28	2,233.80
Applied to removed ratio	1.39	1.25	1.58	2.75

Fresh water applied	
68,095,229.00	*gallons*
2,507.72	*acre-inches*
35.82	*inches/acre*

Process wastewater applied	
7,976,475.00	*gallons*
293.75	*acre-inches*
4.20	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure

Sample date: 05/31/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,700.00	2,800.00	8,300.00							
DL	100.00	100.00	100.00							

Solid Manure

Sample and source description: Solid Manure

Sample date: 10/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,200.00	4,200.00	43,400.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	57.80	21.20	0.00	0.50	14.70	101.00								3,340.00	1,800
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	20.20	13.50	0.00	0.20	6.92	47.80								1,370.00	800
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	139.00	112.00	0.00	0.30	25.80	171.00								4,330.00	2,220
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	21.50	16.50	0.00	7.20	9.00	61.00								1,950.00	1,160
DL	1.00	0.50	0.10	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - DW 1

Domestic Well - DW 1

DW 1

Sample description: DW 1

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	7.30	0.00	7.30								1,540.00	
DL	0.10	0.10	0.10								10.00	

Domestic Well - DW 2

DW 2

Sample description: DW 2

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	7.40	0.00	7.40								1,550.00	
DL	0.10	0.10	0.10								10.00	

Domestic Well - DW 3

DW 3

Sample description: DW 3

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	7.40	0.00	7.40								1,540.00	
DL	0.10	0.10	0.10								10.00	

Irrigation Water - Surface Water

Irrigation Water - Surface Water

Irrigation Water - Surface Water

Sample description: Irrigation Water - Surface Water

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.30	0.00	0.30								268.00	172
DL	0.10	0.10	0.01								10.00	10

D. SOIL ANALYSES

1

1

Sample and source description: 1

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			175.00				
DL			2.00				

10

10

Sample and source description: 10

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			42.00				
DL			2.00				

11A

11A

11

Sample and source description: 11

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			330.00				
DL			2.00				

11B

11

Sample and source description: 11

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			330.00				
DL			2.00				

12

12

Sample and source description: 12

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			95.00				
DL			2.00				

2

2

2

Sample and source description: 2

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			100.00				
DL			2.00				

3

3

Sample and source description: 3

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			2.00				

4

4

Sample and source description: 4

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			99.00				
DL			2.00				

5

5

5

Sample and source description: 5

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			124.00				
DL			2.00				

6

6

Sample and source description: 6

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			141.00				
DL			2.00				

7

7

Sample and source description: 7

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			549.00				
DL			2.00				

8

8

8

Sample and source description: 8

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			86.00				
DL			2.00				

9

9

Sample and source description: 9

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			87.00				
DL			2.00				

E. PLANT TISSUE ANALYSES

10 - 12/03/2022: Ryegrass, hay

10

Sample and source description: 10

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,700.00	4,500.00	43,400.00		13.80
DL	100.00	100.00	100.00		0.01

10 - 05/10/2023: Sudangrass, silage

10 - 05/10/2023: Sudangrass, silage

10

Sample and source description: 10

Sample date: 07/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,400.00	2,500.00	20,800.00		7.40
DL	100.00	100.00	100.00		0.01

11A - 12/03/2022: Ryegrass, hay

11A-1

Sample and source description: 11A-1

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,600.00	4,500.00	42,700.00		13.60
DL	100.00	100.00	100.00		0.01

11A-2

Sample and source description: 11A-2

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,600.00	4,400.00	39,000.00		13.70
DL	100.00	100.00	100.00		0.01

11A - 05/18/2023: Sudangrass, silage

11A - 05/18/2023: Sudangrass, silage

11A-1

Sample and source description: 11A-1

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,600.00	2,500.00	25,800.00		10.30
DL	100.00	100.00	100.00		0.01

11A-2

Sample and source description: 11A-2

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,900.00	2,500.00	26,700.00		11.10
DL	100.00	100.00	100.00		0.01

11B - 12/03/2022: Ryegrass, hay

11B-1

Sample and source description: 11B-1

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,300.00	4,600.00	43,200.00		13.60
DL	100.00	100.00	100.00		0.01

11B - 12/03/2022: Ryegrass, hay

11B-2

Sample and source description: 11B-2

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,700.00	4,400.00	39,800.00		12.90
DL	100.00	100.00	100.00		0.01

11B - 05/10/2023: Sudangrass, silage

11B-1

Sample and source description: 11B-1

Sample date: 07/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,500.00	2,800.00	28,100.00		12.50
DL	100.00	100.00	100.00		0.01

11B-2

Sample and source description: 11B-2

Sample date: 07/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,000.00	2,900.00	29,600.00		11.80
DL	100.00	100.00	100.00		0.01

7 - 12/03/2022: Ryegrass, hay

7 - 12/03/2022: Ryegrass, hay

7

Sample and source description: 7

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,100.00	4,500.00	40,400.00		14.30
DL	100.00	100.00	100.00		0.01

7 - 05/31/2023: Corn, silage

7

Sample and source description: 7

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,600.00	2,500.00	17,100.00		7.00
DL	100.00	100.00	100.00		0.01

8 - 12/03/2022: Ryegrass, hay

8-A

Sample and source description: 8-A

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,000.00	4,600.00	42,500.00		13.20
DL	100.00	100.00	100.00		0.01

8 - 12/03/2022: Ryegrass, hay

8-B

Sample and source description: 8-B

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,600.00	4,800.00	43,700.00		13.10
DL	100.00	100.00	100.00		0.01

8 - 05/31/2023: Corn, silage

8-A

Sample and source description: 8-A

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	2,500.00	17,400.00		7.00
DL	100.00	100.00	100.00		0.01

8-B

Sample and source description: 8-B

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,700.00	2,200.00	16,000.00		6.40
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

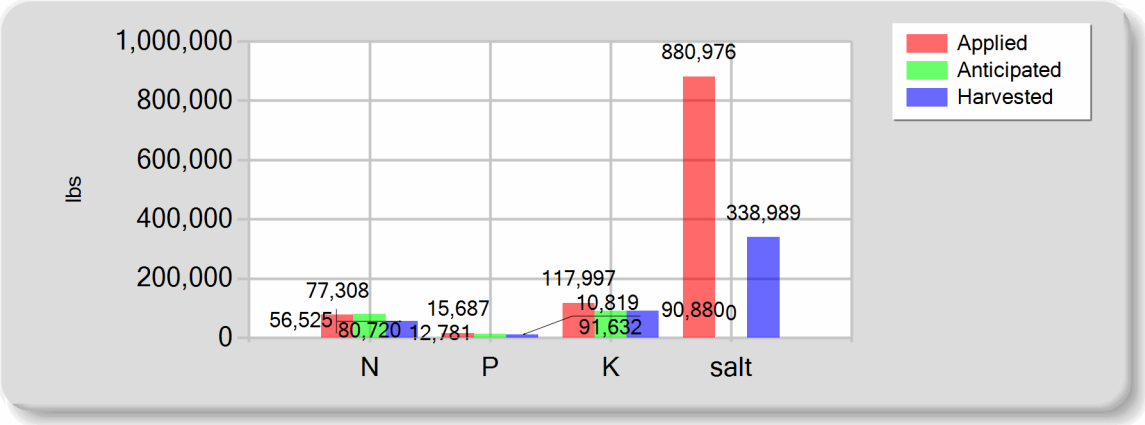
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

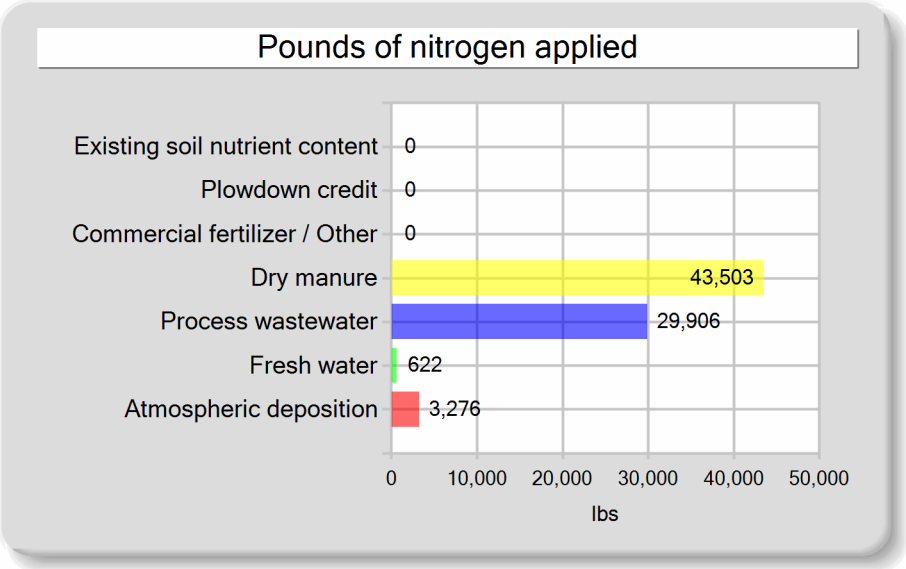
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

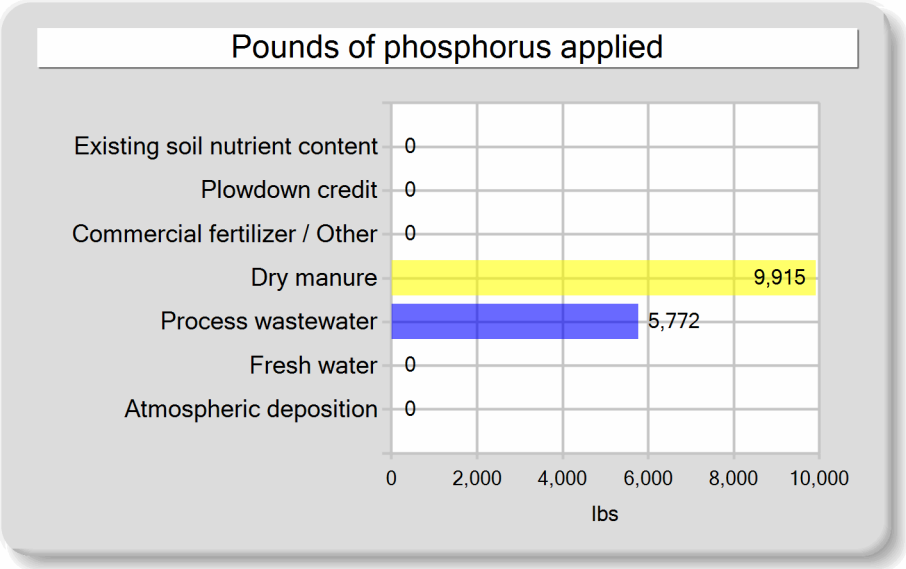
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	43,503.44	9,914.74	79,545.23	0.00
Process wastewater	29,906.14	5,771.95	38,451.56	524,386.03
Fresh water	621.96	0.00	0.00	356,589.88
Atmospheric deposition	3,276.00	0.00	0.00	0.00
Total nutrients applied	77,307.54	15,686.69	117,996.79	880,975.90
Anticipated crop nutrient removal	80,720.00	12,780.80	91,632.00	0.00
Actual crop nutrient removal	56,524.66	10,818.84	90,880.09	338,988.90
Nutrient balance	20,782.88	4,867.85	27,116.71	541,987.00
Applied to removed ratio	1.37	1.45	1.30	2.60

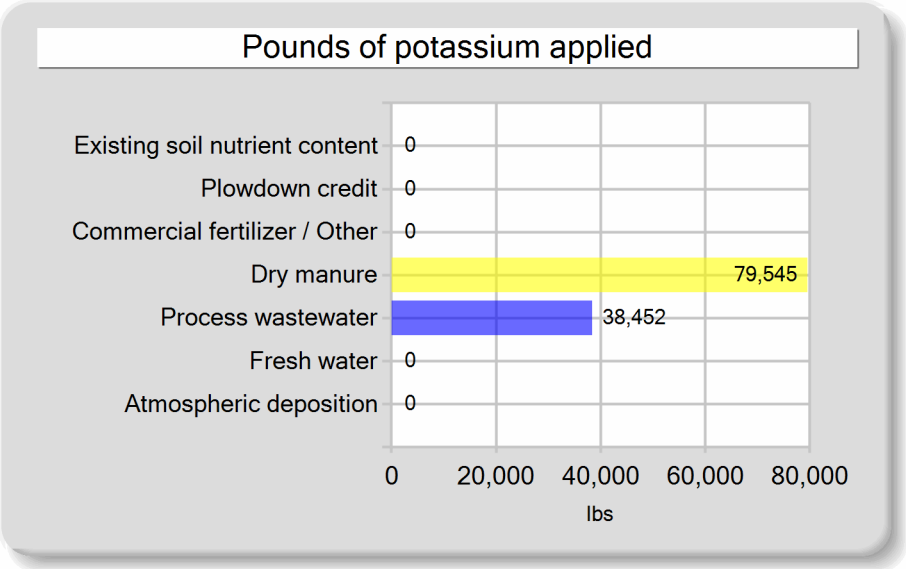
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

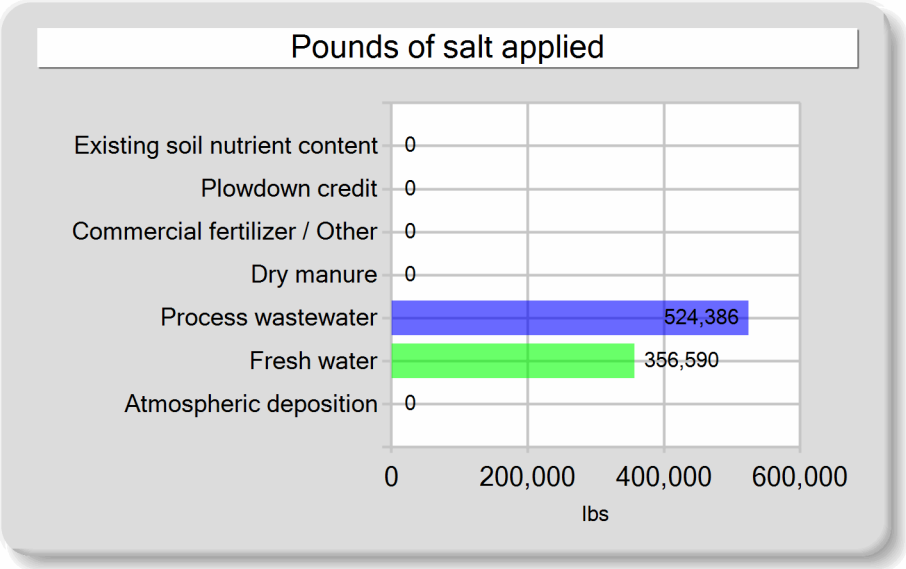


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Ryegrass was harvested as SILAGE instead of HAY

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Lloyd Fagundes
Lloyd Fagundes (Jun 25, 2024 17:05 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Forebay Farms C/O Ralph, Lloyd, &/or Fred Fagundes	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/25/2024	
DATE	DATE

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Forebay Farms

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:08

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H0569-01	DW 1	Ag Water	T. Madruga		08/02/2023 16:00
23H0569-02	DW 2	Ag Water	T. Madruga		08/02/2023 16:00
23H0569-03	DW 3	Ag Water	T. Madruga		08/02/2023 16:00

Default Cooler Temperature on Receipt °C: -1.3
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Sample Results

Sample: DW 1
23H0569-01 (Water)

Sampled: 8/2/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.54**	mmhos/cm	0.01	1		08/04/23 15:10	SM 2510 B		BEH0247
Electrical Conductivity umhos	**1540**	umhos/cm	10.0	1		08/04/23 15:10	SM 2510 B		BEH0247
Ammonia (as N)	*	mg/L	0.00	1		08/02/23 16:00	Field		BEH0310
Nitrate Nitrogen as NO3N	**7.3**	mg/L	0.1	1	10	08/04/23 19:03	EPA 300.0		BEH0235
pH	**7.5**	units	1.0	1		08/04/23 15:10	SM 4500-H+	H	BEH0247
Temperature	**25.0**	°C	0.0	1		08/04/23 15:10	SM 2510 B		BEH0247

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Sample Results
(Continued)

Sample: DW 2
23H0569-02 (Water)

Sampled: 8/2/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.55**	mmhos/cm	0.01	1		08/04/23 15:11	SM 2510 B		BEH0247
Electrical Conductivity umhos	**1550**	umhos/cm	10.0	1		08/04/23 15:11	SM 2510 B		BEH0247
Ammonia (as N)	*	mg/L	0.00	1		08/02/23 16:00	Field		BEH0310
Nitrate Nitrogen as NO3N	**7.4**	mg/L	0.1	1	10	08/04/23 19:30	EPA 300.0		BEH0235
pH	**7.5**	units	1.0	1		08/04/23 15:11	SM 4500-H+	H	BEH0247
Temperature	**25.0**	°C	0.0	1		08/04/23 15:11	SM 2510 B		BEH0247

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Ranch: Forebay Farms

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:08

Sample Results
(Continued)

Sample: DW 3
23H0569-03 (Water)

Sampled: 8/2/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.54**	mmhos/cm	0.01	1		08/04/23 15:12	SM 2510 B		BEH0247
Electrical Conductivity umhos	**1540**	umhos/cm	10.0	1		08/04/23 15:12	SM 2510 B		BEH0247
Ammonia (as N)	*	mg/L	0.00	1		08/02/23 16:00	Field		BEH0310
Nitrate Nitrogen as NO3N	**7.4**	mg/L	0.1	1	10	08/04/23 19:50	EPA 300.0		BEH0235
pH	**7.5**	units	1.0	1		08/04/23 15:12	SM 4500-H+	H	BEH0247
Temperature	**25.0**	°C	0.0	1		08/04/23 15:12	SM 2510 B		BEH0247

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Received: 08/03/2023 15:10
Reported: 08/07/2023 12:08

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0235										
Blank (BEH0235-BLK1)					Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0235-BLK2)					Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEH0235-BS1)					Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
Duplicate (BEH0235-DUP1)		**Source: 23H0572-01**			Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	2.9		0.1	mg/L		2.9			0.988	10
Matrix Spike (BEH0235-MS1)		**Source: 23H0572-01**			Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	8.0		0.1	mg/L	5.000	2.9	101	90-110		
Reference (BEH0235-SRM1)					Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEH0235-SRM2)					Prepared: 8/4/2023 Analyzed: 8/5/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		

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Ranch: Forebay Farms

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:08

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0247										
Blank (BEH0247-BLK1)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.4		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0247-BLK2)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.5		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0247-BLK3)					Prepared & Analyzed: 8/4/2023					
pH	7.0		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Duplicate (BEH0247-DUP1)		**Source: 23H0569-01**			Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1.52		0.01	mmhos/cm		1.54			1.70	10
pH	7.5		1.0	units		7.5			0.133	10
Electrical Conductivity umhos	1520		10.0	umhos/cm		1540			1.70	10
Duplicate (BEH0247-DUP2)		**Source: 23H0612-01**			Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	0.14		0.01	mmhos/cm		0.14			0.0713	10
pH	7.8		1.0	units		7.6			1.82	10
Electrical Conductivity umhos	140		10.0	umhos/cm		140			0.0712	10
Reference (BEH0247-SRM1)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	544			umhos/cm	538.0		101	90-110		
Reference (BEH0247-SRM2)					Prepared & Analyzed: 8/4/2023					
pH	7.9			units	7.790		101	.7163-101.28		
Reference (BEH0247-SRM3)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEH0247-SRM4)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEH0247-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0247 (Continued)										
Reference (BEH0247-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEH0247-SRM6)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH0247-SRM7)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEH0247-SRM8)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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08/03/23 15:10 23H0569

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1	1							
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										

59

08/03/23 15:10 23H0569

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples 3 No. of Bottles: 3

Purchase Order No | Bill To: | Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:
REQUESTED BY: Rui & John Avila
PROJECT Forebay Farms
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) **DWW1: EC, pH, NO_3-N (NH4-N Field Test)**
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	DW 1	8/2/2023	1600	-1.3	
2	DW 2	8/2/2023	1600	-0.3	
3	DW 3	8/2/2023	1600	-1.6	
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	8/2/2023 @ 1600	8/3/23 @ 0900
Second	JCF	Dellavalle Laboratory, Inc.	8/3/23 @ 0900	8-3-23 2:30
Third				
Fourth	[illegible]	DLI	8/3/23 15:10	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

08/03/23 15:10 23H0569

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1	1							
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Forebay Farms

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:08

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H0569-01	DW 1	Ag Water	T. Madruga		08/02/2023 16:00
23H0569-02	DW 2	Ag Water	T. Madruga		08/02/2023 16:00
23H0569-03	DW 3	Ag Water	T. Madruga		08/02/2023 16:00

Default Cooler Temperature on Receipt °C: -1.3
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Forebay Farms

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:08

Sample Results

Sample: DW 1
23H0569-01 (Water)

Sampled: 8/2/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.54**	mmhos/cm	0.01	1		08/04/23 15:10	SM 2510 B		BEH0247
Electrical Conductivity umhos	**1540**	umhos/cm	10.0	1		08/04/23 15:10	SM 2510 B		BEH0247
Ammonia (as N)	*	mg/L	0.00	1		08/02/23 16:00	Field		BEH0310
Nitrate Nitrogen as NO3N	**7.3**	mg/L	0.1	1	10	08/04/23 19:03	EPA 300.0		BEH0235
pH	**7.5**	units	1.0	1		08/04/23 15:10	SM 4500-H+	H	BEH0247
Temperature	**25.0**	°C	0.0	1		08/04/23 15:10	SM 2510 B		BEH0247

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Forebay Farms

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:08

Sample Results
(Continued)

Sample: DW 2
23H0569-02 (Water)

Sampled: 8/2/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.55**	mmhos/cm	0.01	1		08/04/23 15:11	SM 2510 B		BEH0247
Electrical Conductivity umhos	**1550**	umhos/cm	10.0	1		08/04/23 15:11	SM 2510 B		BEH0247
Ammonia (as N)	*	mg/L	0.00	1		08/02/23 16:00	Field		BEH0310
Nitrate Nitrogen as NO3N	**7.4**	mg/L	0.1	1	10	08/04/23 19:30	EPA 300.0		BEH0235
pH	**7.5**	units	1.0	1		08/04/23 15:11	SM 4500-H+	H	BEH0247
Temperature	**25.0**	°C	0.0	1		08/04/23 15:11	SM 2510 B		BEH0247

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Forebay Farms

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:08

Sample Results
(Continued)

Sample: DW 3
23H0569-03 (Water)

Sampled: 8/2/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.54**	mmhos/cm	0.01	1		08/04/23 15:12	SM 2510 B		BEH0247
Electrical Conductivity umhos	**1540**	umhos/cm	10.0	1		08/04/23 15:12	SM 2510 B		BEH0247
Ammonia (as N)	*	mg/L	0.00	1		08/02/23 16:00	Field		BEH0310
Nitrate Nitrogen as NO3N	**7.4**	mg/L	0.1	1	10	08/04/23 19:50	EPA 300.0		BEH0235
pH	**7.5**	units	1.0	1		08/04/23 15:12	SM 4500-H+	H	BEH0247
Temperature	**25.0**	°C	0.0	1		08/04/23 15:12	SM 2510 B		BEH0247

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Forebay Farms

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:08

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0235										
Blank (BEH0235-BLK1)					Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0235-BLK2)					Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEH0235-BS1)					Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
Duplicate (BEH0235-DUP1)		**Source: 23H0572-01**			Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	2.9		0.1	mg/L		2.9			0.988	10
Matrix Spike (BEH0235-MS1)		**Source: 23H0572-01**			Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	8.0		0.1	mg/L	5.000	2.9	101	90-110		
Reference (BEH0235-SRM1)					Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEH0235-SRM2)					Prepared: 8/4/2023 Analyzed: 8/5/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Forebay Farms

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:08

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0247										
Blank (BEH0247-BLK1)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.4		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0247-BLK2)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.5		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0247-BLK3)					Prepared & Analyzed: 8/4/2023					
pH	7.0		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Duplicate (BEH0247-DUP1)		**Source: 23H0569-01**			Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1.52		0.01	mmhos/cm		1.54			1.70	10
pH	7.5		1.0	units		7.5			0.133	10
Electrical Conductivity umhos	1520		10.0	umhos/cm		1540			1.70	10
Duplicate (BEH0247-DUP2)		**Source: 23H0612-01**			Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	0.14		0.01	mmhos/cm		0.14			0.0713	10
pH	7.8		1.0	units		7.6			1.82	10
Electrical Conductivity umhos	140		10.0	umhos/cm		140			0.0712	10
Reference (BEH0247-SRM1)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	544			umhos/cm	538.0		101	90-110		
Reference (BEH0247-SRM2)					Prepared & Analyzed: 8/4/2023					
pH	7.9			units	7.790		101	.7163-101.28		
Reference (BEH0247-SRM3)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEH0247-SRM4)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEH0247-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		

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LABORATORY INC

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Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Forebay Farms

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:08

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0247 (Continued)										
Reference (BEH0247-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEH0247-SRM6)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH0247-SRM7)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEH0247-SRM8)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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08/03/23 15:10 23H0569

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1	1							
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										

59

08/03/23 15:10 23H0569

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples 3 No. of Bottles: 3

Purchase Order No | Bill To: | Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:
REQUESTED BY: Rui & John Avila
PROJECT Forebay Farms
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) **DWW1: EC, pH, NO_3-N (NH4-N Field Test)**
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	DW 1	8/2/2023	1600	-1.3	
2	DW 2	8/2/2023	1600	-0.3	
3	DW 3	8/2/2023	1600	-1.6	
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	8/2/2023 @ 1600	8/3/23 @ 0900
Second	JCF	Dellavalle Laboratory, Inc.	8/3/23 @ 0900	8-3-23 2:30
Third				
Fourth	[illegible]	DLI	8/3/23 15:10	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
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Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

08/03/23 15:10 23H0569

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1	1							
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Forebay Farms

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:08

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H0569-01	DW 1	Ag Water	T. Madruga		08/02/2023 16:00
23H0569-02	DW 2	Ag Water	T. Madruga		08/02/2023 16:00
23H0569-03	DW 3	Ag Water	T. Madruga		08/02/2023 16:00

Default Cooler Temperature on Receipt °C: -1.3
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Ranch: Forebay Farms

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:08

Sample Results

Sample: DW 1
23H0569-01 (Water)

Sampled: 8/2/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.54**	mmhos/cm	0.01	1		08/04/23 15:10	SM 2510 B		BEH0247
Electrical Conductivity umhos	**1540**	umhos/cm	10.0	1		08/04/23 15:10	SM 2510 B		BEH0247
Ammonia (as N)	*	mg/L	0.00	1		08/02/23 16:00	Field		BEH0310
Nitrate Nitrogen as NO3N	**7.3**	mg/L	0.1	1	10	08/04/23 19:03	EPA 300.0		BEH0235
pH	**7.5**	units	1.0	1		08/04/23 15:10	SM 4500-H+	H	BEH0247
Temperature	**25.0**	°C	0.0	1		08/04/23 15:10	SM 2510 B		BEH0247

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Received: 08/03/2023 15:10
Reported: 08/07/2023 12:08

Sample Results
(Continued)

Sample: DW 2
23H0569-02 (Water)

Sampled: 8/2/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.55**	mmhos/cm	0.01	1		08/04/23 15:11	SM 2510 B		BEH0247
Electrical Conductivity umhos	**1550**	umhos/cm	10.0	1		08/04/23 15:11	SM 2510 B		BEH0247
Ammonia (as N)	*	mg/L	0.00	1		08/02/23 16:00	Field		BEH0310
Nitrate Nitrogen as NO3N	**7.4**	mg/L	0.1	1	10	08/04/23 19:30	EPA 300.0		BEH0235
pH	**7.5**	units	1.0	1		08/04/23 15:11	SM 4500-H+	H	BEH0247
Temperature	**25.0**	°C	0.0	1		08/04/23 15:11	SM 2510 B		BEH0247

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Ranch: Forebay Farms

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:08

Sample Results
(Continued)

Sample: DW 3
23H0569-03 (Water)

Sampled: 8/2/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.54**	mmhos/cm	0.01	1		08/04/23 15:12	SM 2510 B		BEH0247
Electrical Conductivity umhos	**1540**	umhos/cm	10.0	1		08/04/23 15:12	SM 2510 B		BEH0247
Ammonia (as N)	*	mg/L	0.00	1		08/02/23 16:00	Field		BEH0310
Nitrate Nitrogen as NO3N	**7.4**	mg/L	0.1	1	10	08/04/23 19:50	EPA 300.0		BEH0235
pH	**7.5**	units	1.0	1		08/04/23 15:12	SM 4500-H+	H	BEH0247
Temperature	**25.0**	°C	0.0	1		08/04/23 15:12	SM 2510 B		BEH0247

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Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Forebay Farms

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:08

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0235										
Blank (BEH0235-BLK1)					Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0235-BLK2)					Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEH0235-BS1)					Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
Duplicate (BEH0235-DUP1)		**Source: 23H0572-01**			Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	2.9		0.1	mg/L		2.9			0.988	10
Matrix Spike (BEH0235-MS1)		**Source: 23H0572-01**			Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	8.0		0.1	mg/L	5.000	2.9	101	90-110		
Reference (BEH0235-SRM1)					Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEH0235-SRM2)					Prepared: 8/4/2023 Analyzed: 8/5/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		

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Ranch: Forebay Farms

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:08

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0247										
Blank (BEH0247-BLK1)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.4		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0247-BLK2)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.5		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0247-BLK3)					Prepared & Analyzed: 8/4/2023					
pH	7.0		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Duplicate (BEH0247-DUP1)		**Source: 23H0569-01**			Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1.52		0.01	mmhos/cm		1.54			1.70	10
pH	7.5		1.0	units		7.5			0.133	10
Electrical Conductivity umhos	1520		10.0	umhos/cm		1540			1.70	10
Duplicate (BEH0247-DUP2)		**Source: 23H0612-01**			Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	0.14		0.01	mmhos/cm		0.14			0.0713	10
pH	7.8		1.0	units		7.6			1.82	10
Electrical Conductivity umhos	140		10.0	umhos/cm		140			0.0712	10
Reference (BEH0247-SRM1)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	544			umhos/cm	538.0		101	90-110		
Reference (BEH0247-SRM2)					Prepared & Analyzed: 8/4/2023					
pH	7.9			units	7.790		101	.7163-101.28		
Reference (BEH0247-SRM3)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEH0247-SRM4)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEH0247-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		

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Received: 08/03/2023 15:10
Reported: 08/07/2023 12:08

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0247 (Continued)										
Reference (BEH0247-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEH0247-SRM6)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH0247-SRM7)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEH0247-SRM8)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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08/03/23 15:10 23H0569

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1	1							
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										

59

08/03/23 15:10 23H0569

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples 3 No. of Bottles: 3

Purchase Order No | Bill To: | Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:
REQUESTED BY: Rui & John Avila
PROJECT Forebay Farms
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) **DWW1: EC, pH, NO_3-N (NH4-N Field Test)**
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	DW 1	8/2/2023	1600	-1.3	
2	DW 2	8/2/2023	1600	-0.3	
3	DW 3	8/2/2023	1600	-1.6	
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	8/2/2023 @ 1600	8/3/23 @ 0900
Second	JCF	Dellavalle Laboratory, Inc.	8/3/23 @ 0900	8-3-23 2:30
Third				
Fourth	[illegible]	DLI	8/3/23 15:10	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

08/03/23 15:10 23H0569

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1	1							
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										